Exp 10 · JFET common-source amplifier: frequency response

Manual pp. 29–30 · LTspice · CO4, CO5 · the JFET is the 2N5434 (the manual's "2N54349" is a typo)

9 JFET curves → **10 CS amplifier** → 11 Oscillator → 12 MOSFET inverter

What you already have: the BJT's CE amplifier (Exp 6), meaning bias, coupling and bypass capacitors, gain, and −3 dB points. The JFET's g_m (Exp 9).

Two terms first:

- **Common source (CS):** the FET's version of CE. Input at the gate, output at the drain, source common.
- **Self-bias:** the source resistor R_s lifts the source above the gate, which makes V_{GS} negative without a negative supply.

MANUAL EXP 10, AIM

To plot the frequency response of a common-source amplifier. Viva form: "Estimate the gain from g_m and the resistors, and say why the output is 180° out of phase."

GUESS FIRST

CE gain was R_C/r_e. What do you think replaces $1/r_e$ for a FET?

The circuit and its bias

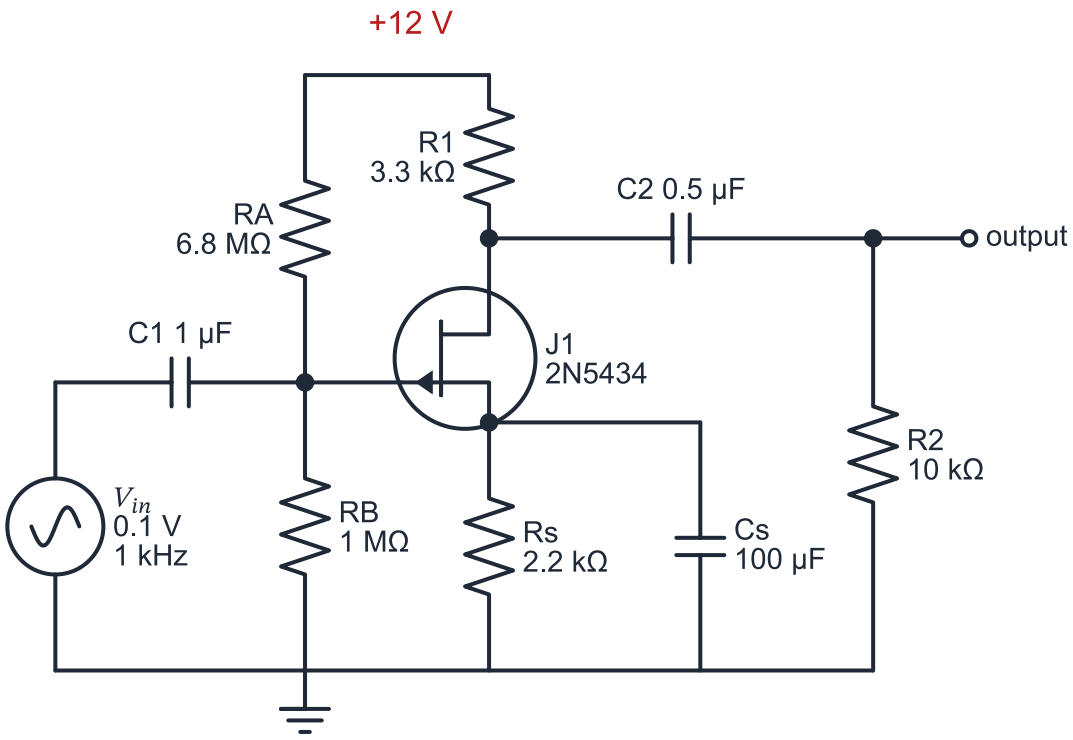


Fig 10.1, redrawn: gate divider RA/RB, drain resistor R1, source resistor Rs bypassed by Cs, 10 kΩ load through C2.

Part	Job
RA 6.8 MΩ, RB 1 MΩ	gate divider: $V_G = 12 \times 1/7.8 \approx 1.54$ V. Megohms are fine because the gate draws no current.
Rs 2.2 kΩ	self-bias: $V_{GS} = V_G - I_D R_s$ is negative
Cs 100 µF	bypasses Rs for AC, giving full gain (as C_E did in Exp 6)
R1 3.3 kΩ	drain load: converts the I_D change into a voltage change
C1, C2	coupling: block DC, pass the signal

Q-point (LTspice .op): $V_G = 1.54$ V, $V_S = 3.17$ V, so **$V_{GS} = -1.63$ V**. $I_D = 3.17/2.2\text{k} =$ **1.44 mA**, and $V_D = 7.25$ V. The drain sits well above the source, so the JFET is in saturation ✓.

Gain by hand: the small-signal model is a current source $g_m v_{gs}$ into the drain.

$$A_v = -g_m (R_1 \parallel R_2) = -g_m \times (3.3\ \text{k}\Omega \parallel 10\ \text{k}\Omega) = -g_m \times 2.48\ \text{k}\Omega$$

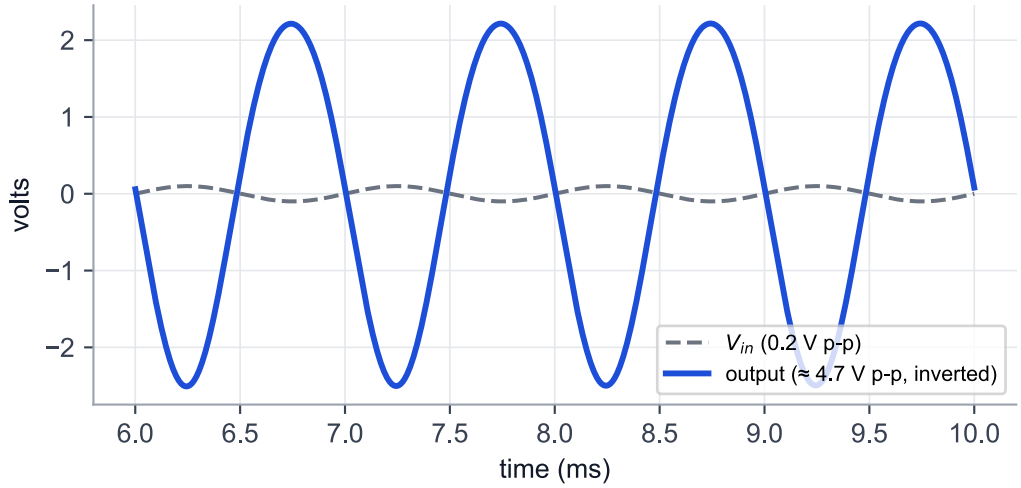
For the 2N5434 at this Q-point, $g_m \approx 9.7$ mS (from its model: $I_{DSS} \approx 65$ mA, $V_P \approx -1.9$ V, using $g_m = (2I_{DSS}/|V_P|)(1 - V_{GS}/V_P)$). So $A_v \approx -24$. The **minus sign** is the 180° phase inversion: as v_{gs} goes up, I_D goes up, and the drain voltage $V_{DD} - I_D R_1$ goes **down**.

Do it

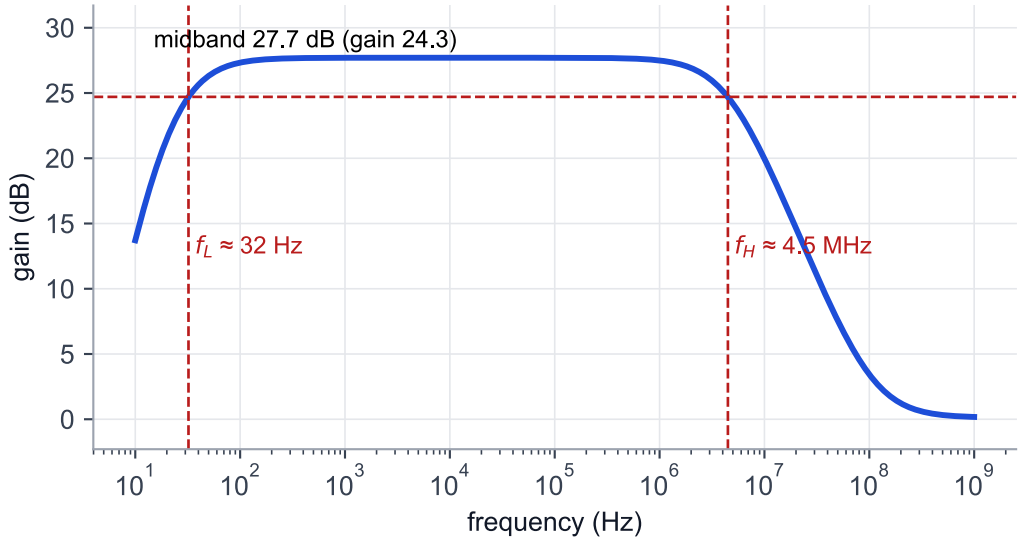
DO IT (LTSPICE)

1. Draw Fig 10.1 with J1 = **2N5434**, VDD = 12 V. Name the output node `out` (across R2).
2. Vin: `SINE(0 0.1 1k)` with AC amplitude `1`.
3. **Transient:** `.tran 10m`. Plot the input and V(out), and read both peak-to-peak values with cursors.
4. **Frequency response:** `.ac dec 10 100 100Meg`. Read the midband dB and find the two −3 dB frequencies.
5. Fill the table at 100 Hz, 1k, 5k, 10k, 50k, 100k, 500k, 1M, 5M, 10M, 100M.

What you should see



Transient: 0.2 V p-p in, ≈ 4.7 V p-p out, inverted.



AC: midband gain 24.3 (27.7 dB), f_L ≈ 32 Hz, f_H ≈ 4.5 MHz.

f (Hz)	100	1k	5k	10k	50k	100k	500k	1M	5M	10M	100M
gain	23.3	24.3	24.3	24.3	24.3	24.3	24.1	23.7	16.2	10.0	1.5
dB	27.3	27.7	27.7	27.7	27.7	27.7	27.6	27.5	24.2	20.0	3.4

Calculation

- Transient gain = $V_{O(pp)}/V_{in(pp)}$ = 4.73/0.2 = **23.6**.
- AC midband gain = **24.3**.
- BW = $f_H - f_L$ = 4.48 MHz − 32 Hz ≈ **4.5 MHz**.

The low cut-off comes from Cs (with $R_s \parallel 1/g_m \approx 100\ \Omega$) and from C2 (with R1 + R2). The high cut-off comes from the JFET's gate–drain and gate–source capacitances (35 pF each in its model), Miller-multiplied.

WRITE IN YOUR RECORD

Result: maximum gain ≈ **24.3 (27.7 dB)**; transient gain 4.73 V/0.2 V = 23.6; output 180° out of phase; f_L ≈ 32 Hz, f_H ≈ 4.5 MHz, bandwidth ≈ **4.5 MHz**.

Theory for the record: in a common-source amplifier the signal is applied to the gate and taken from the drain, with the source common (bypassed to AC ground). The gate divider and source resistor set a negative V_{GS} that biases the JFET in saturation. A gate-voltage change produces a drain-current change $g_m v_{gs}$, which develops an output voltage across the drain load: $A_v = -g_m(R_D \parallel R_L)$. The output is inverted (180°). The input resistance is very high (set by the gate divider). The gain is lower than a BJT CE stage, but the input resistance and noise performance are better. Coupling/bypass capacitors set f_L; device capacitances set f_H.

CHECK 1

(a) Remove Cs. What happens to the gain, and why? (b) Why can the gate resistors be megohms here, when the BJT's bias resistors were kilohms? (c) Estimate the input resistance seen by the source.

Answers

Opening question. g_m replaces $1/r_e$: $A_v = -g_m(R_D \parallel R_L) \approx -24$. The output is inverted because a rising gate voltage raises I_D, which increases the drop across R1 and lowers the drain voltage.

Check 1. (a) The gain drops: R_s now gives negative feedback (as in Exp 8). $A_v \approx -g_m R_D'/(1 + g_m R_s) = -24/(1 + 9.7\text{m} \times 2.2\text{k}) \approx$ **−1.1**. (b) The JFET gate takes no current, so the divider is not loaded. A BJT's base current would make megohm resistors drop volts and wreck the bias. (c) $R_{in} \approx RA \parallel RB = 6.8\ \text{M}\Omega \parallel 1\ \text{M}\Omega \approx$ **0.87 MΩ**.